SIMATS ENGINEERING
ASSESSMENT TOOL 1: Constraint-Based Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED
CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 50**
Question Count: 5

Code of Conduct
I Bala S (192524427) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: bala

Assessment Tasks
Constraint-Based Problem Solving
1. Design an HDFS-based storage solution for a media platform under the constraints that data availability must remain above 99.9%, replication factor cannot exceed 3, and storage growth must stay cost efficient. Justify your design.
2. A MapReduce workflow must process log data within a 20-minute batch window using limited cluster nodes. Propose a job design under these constraints and explain task distribution.
3. A YARN-managed cluster supports ETL, reporting, and machine learning jobs. Design a scheduler policy under the constraints of fairness, priority for ETL, and recovery from node failure.
4. An enterprise needs to ingest data from SAN, NAS, and operational databases into a big data platform while maintaining backup reliability and minimal duplication. Propose a constrained architecture and justify each component.
5. Design an end-to-end Hadoop ecosystem solution under the constraints of secure data movement, high availability, and integration with Apache NiFi or ETL pipelines. Explain how your design satisfies all constraints.

Constraint-Based Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Constraints	25%	All technical constraints are identified and prioritized accurately	Most constraints identified	Some constraints identified	Constraints poorly understood
System Design Quality	25%	Design is robust, feasible, and aligned to constraints	Reasonable design	Basic design with gaps	Weak design
Application of Hadoop Concepts	20%	Uses HDFS, MapReduce, YARN, and integration concepts effectively	Mostly effective use	Partial use of concepts	Weak concept application
Justification	15%	Strong technical reasoning for all choices	Good justification	Basic justification	Little justification
Clarity	15%	Clear, well-structured, and professional answer	Mostly clear	Somewhat unclear	Poorly structured

1.Design an HDFS-based storage solution for a media platform under the constraints that data availability must remain above 99.9%, replication factor cannot exceed 3, and storage growth must stay cost efficient. Justify your design.

- Use HDFS as the main distributed storage system.
- Set replication factor = 3 (maximum allowed).
- Divide large media files into HDFS blocks, e.g., 128/256 MB blocks.
- Store replicas on different Data Nodes and racks using rack awareness.
- Use Name Node High Availability with Active and Standby Name Nodes.
- Add Data Nodes as storage grows instead of replacing the entire cluster.
- Use cold/archive storage for old media to reduce cost.

Replication factor = 3

Required availability = 99.9%

Assume each Data Node has availability of 99%.

Probability that all 3 replicas fail:

P(failure) = (1−0.99)3 = 0.013 = 0.000001

Availability = 1−0.000001

=0.999999=99.9999%

Result: 99.9999% > 99.9%, so the design satisfies the availability constraint.

2.A MapReduce workflow must process log data within a 20-minute batch window using limited cluster nodes. Propose a job design under these constraints and explain task distribution.

MapReduce workflow for a 20-minute batch window

- Split logs into multiple HDFS blocks.
- Assign one mapper to each input split.
- Run reducers in parallel on available nodes.
- Give higher parallelism to the mapper stage because log data is usually large.
- Use the limited nodes efficiently by reusing nodes for multiple tasks.

20-minute constraint:

If there are N input blocks and M mapper slots:

Mapper waves $\approx N/M$

Choose the number of reducers and mapper slots so that:

Input + Map + Shuffle + Reduce < 20 minutes

- Log data = 600 GB
- HDFS block size = 128 MB
- Mapper processing speed = 100 MB/s
- Available mapper slots = 10

Number of blocks:

$$600GB = 600 \times 1024 = 614400MB$$

$$Blocks = 614400/128 = 4800$$

Mapper waves:

$$4800/10 = 480 \text{ waves}$$

At 100 MB/s, each 128 MB block takes:

$$128/100=1.28 \text{ sec}$$

Approximate map time:

$$480\times1.28=614.4 \text{ sec} =10.24 \text{ minutes}$$

Adding shuffle and reduce time, suppose:

$$10.24+6=16.24 \text{ minutes}$$

Result: 16.24 < 20 minutes, so the job satisfies the batch-window constraint.

3.A YARN-managed cluster supports ETL, reporting, and machine learning jobs. Design a scheduler policy under the constraints of fairness, priority for ETL, and recovery from node failure

YARN scheduler policy

- YARN Node Manager detects failed nodes through heartbeats.
- Failed tasks are automatically rescheduled on healthy nodes.
- Application Master recovery allows applications to restart.
- HDFS replicas ensure required input data remains available.

Suppose the cluster has 100 GB RAM.

Allocation:

- ETL = 50%
- Reporting = 30%
- ML = 20%

Therefore:

$$ETL = 100 \times 0.50 = 50GB$$

$$Reporting = 100 \times 0.30 = 30GB$$

$$ML = 100 \times 0.20 = 20GB$$

Total:

$$50 + 30 + 20 = 100GB$$

Result: All cluster resources are allocated while ETL receives the highest priority.

4.An enterprise needs to ingest data from SAN, NAS, and operational databases into a big data platform while maintaining backup reliability and minimal duplication. Propose a constrained architecture and justify each component.

Apache NiFi

- Collects files from SAN/NAS.
- Controls data flow and provides provenance.
- Supports routing, transformation and monitoring.

Database ingestion

- Use JDBC-based ingestion/ETL pipelines.
- Perform incremental extraction using timestamps or transaction IDs instead of repeatedly copying the entire database.

HDFS

- Central distributed storage.
- Replication factor 3 provides fault tolerance.

SAN data = 2 TB

NAS data = 3 TB

Database data = 1 TB

Total input:

$$2+3+1=6TB$$

Suppose 10% is duplicate data:

$$Duplicate=6\times0.10=0.6TB$$

Unique data:

$$6-0.6=5.4TB$$

With HDFS replication factor 3:

$\text{Storage} = 5.4 \times 3 = 16.2\text{TB}$

Result: About 16.2 TB physical HDFS storage is required after deduplication.

5.Design an end-to-end Hadoop ecosystem solution under the constraints of secure data movement, high availability, and integration with Apache NiFi or ETL pipelines. Explain how your design satisfies all constraints.

1.Data ingestion

- Apache NiFi collects files, database records and streaming data.
- Use processors for filtering, transformation and routing.

2. Secure data movement

- Use TLS/SSL for data transmission.
- Enable Kerberos authentication for Hadoop services.
- Apply HDFS permissions and access control.
- Encrypt sensitive data where required.

3. High availability

- Use Name Node HA with Active and Standby Name Nodes.

- Use multiple Data Nodes.
- Replication factor of 3 protects against node failures.
- Use YARN Resource Manager HA for cluster management.

- Daily incoming data = **500 GB**
- Retention = **30 days**
- HDFS replication = **3**

Logical storage:

$$500 \times 30 = 15000GB = 15TB$$

Physical storage with replication:

$$15 \times 3 = 45TB$$

Add 20% extra capacity for growth:

$$45 \times 1.20 = 54TB$$

Result: The cluster should provide approximately 54 TB of physical storage.